Title: Whole Genome Sequencing Reanalysis of a High-Ethanol-Producing *Saccharomyces cerevisiae* Mutant Strain

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Abstract

Whole Genome Sequencing (WGS) data from wild-type and high-ethanol-producing mutant strains of *Saccharomyces cerevisiae* were reanalyzed to identify genetic variants specific to the mutant strain and candidate genes potentially related to higher ethanol production. Public paired-end sequencing reads were quality checked, trimmed, mapped to the S. cerevisiae R64 reference genome, and analyzed through variant calling, comparison, annotation, and prioritization using an alternative bioinformatics workflow. The analysis identified 36,871 mutant-specific variants, including effects classified as HIGH and MODERATE impact. Comparison with ethanol-related genes from the source study showed that 27 genes overlapped with genes containing mutant-specific variants. Among these genes, MAL31, AAD4, and NTH1 were prioritized as the strongest candidate genes, suggesting potential links between sugar utilization, alcohol and aldehyde metabolism, stress-related carbon reserve, and the higher ethanol production phenotype.

Background

Saccharomyces cerevisiae is an important yeast species in biotechnology and industrial fermentation, widely used as a model eukaryote and known for its central role in ethanol production through the conversion of sugars into alcohol (Parapouli *et al.,* 2020). Lu *et al.* (2023) reported that a mutant strain of *S. cerevisiae* produced a higher ethanol level, reaching 16.13%, compared with 13.72% in the wild-type strain.

Whole Genome Sequencing (WGS) enables genome-wide identification of genetic variants and provides a foundation for biological interpretation (Koboldt, 2020). In the context of mutant and wild-type strain comparison, this approach can be used to identify mutant-specific variants and evaluate their potential biological effects.

This mini project re-analyzes publicly available WGS data from Lu et al. (2023), including wild-type and mutant strains of *S. cerevisiae*, using an alternative variant analysis workflow. The analysis focuses on finding genetic variants that are specific to the mutant strain and understanding which genes may be affected by those changes. The results are then compared with ethanol-related genes and pathways discussed in the original study to explore candidate genetic changes that may be linked to improved fermentation performance and higher ethanol production in the mutant strain.

Methods

1. Data Acquisition

Public WGS data from Lu et al. (2023) were obtained from the NCBI Sequence Read Archive (SRA). Two sequencing runs were used in this analysis: SRR21747981 for the wild-type strain and SRR21747980 for the mutant strain of *S. cerevisiae*. The SRA files were downloaded using SRA Toolkit and converted into paired-end FASTQ files.

2. Read Quality Control and Preprocessing

Raw read quality was assessed using FastQC. Adapter detection and read trimming were performed using fastp.

3. Reference Genome Preparation and Read Mapping

The reference genome GCF_000146045.2_R64 was downloaded from the NCBI genome database and indexed using BWA. Cleaned reads from the wild-type and mutant samples were mapped separately to the reference genome using BWA-MEM, generating SAM alignment files.

4. Alignment Processing

The SAM alignment files were converted into BAM format, sorted, and indexed using SAMtools. The processed BAM files were used as input for variant calling.

5. Variant Calling, Filtering, and Normalization

Variants were called separately for the wild-type and mutant samples using FreeBayes. Low-quality variants were filtered using BCFtools. The filtered variants were then normalized using BCFtools to standardize variant representation before comparison.

6. Identification of Mutant-Specific Variants

The normalized VCF files from the wild-type and mutant samples were compressed and indexed using bgzip and tabix. BCFtools isec was used to compare both VCF files and identify variants found only in the mutant strain. These mutant-specific variants were selected for variant annotation.

7. Variant Annotation

Before annotation, chromosome names in the mutant-specific VCF file were adjusted to match the chromosome naming format used by the SnpEff R64-1-1.99 database. The mutant-specific variants were then annotated using SnpEff. The SnpEff summary statistics were visualized using ngi_visualizations. Key annotation fields, including gene ID, gene name, predicted impact, predicted effect, and HGVS notation, were extracted using SnpSift and saved as a TSV table.

8. Variant Prioritization

The extracted annotation table was processed using a custom Python script to retain only HIGH and MODERATE impact annotations. The filtered results were then compared with ethanol-related genes and pathways discussed in Lu et al. (2023) to identify candidate genes potentially related to higher ethanol production.

Results

Post-trimming quality control showed that both wild-type reads (Figures 1 and 2) and mutant reads (Figures 3 and 4) of *S. cerevisiae* had high per base sequence quality and no detectable adapter content. Therefore, the cleaned reads were considered suitable for the mapping step.

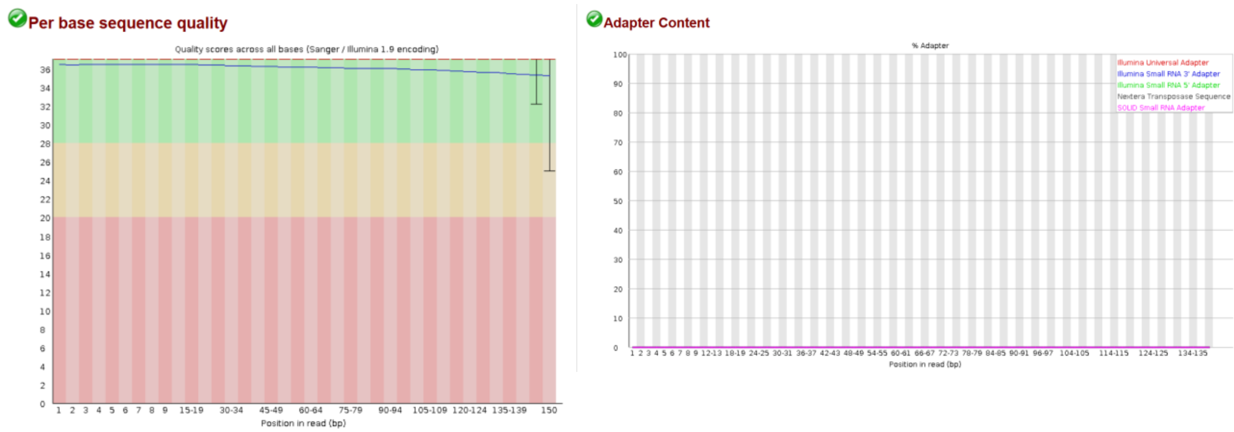

Figure 1. Per-base sequence quality and adapter content of forward reads from the *Saccharomyces cerevisiae* wild-type sample

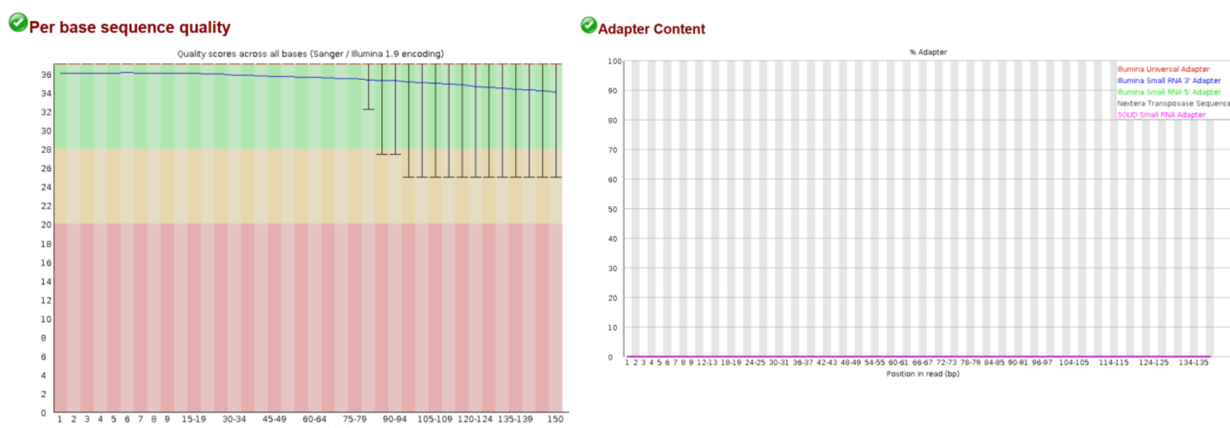

Figure 2. Per-base sequence quality and adapter content of reverse reads from the *Saccharomyces cerevisiae* wild-type sample

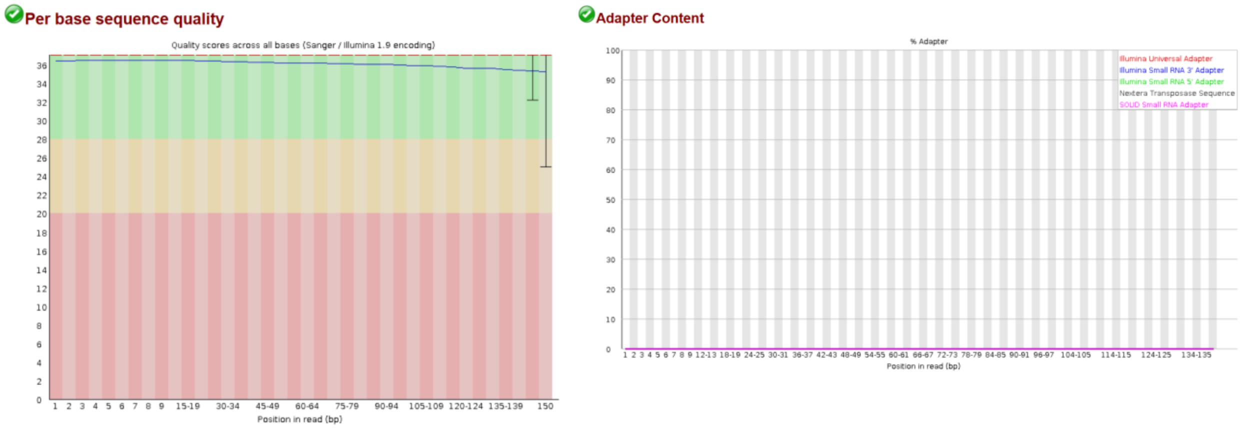

Figure 3. Per-base sequence quality and adapter content of forward reads from the *Saccharomyces cerevisiae* mutant sample

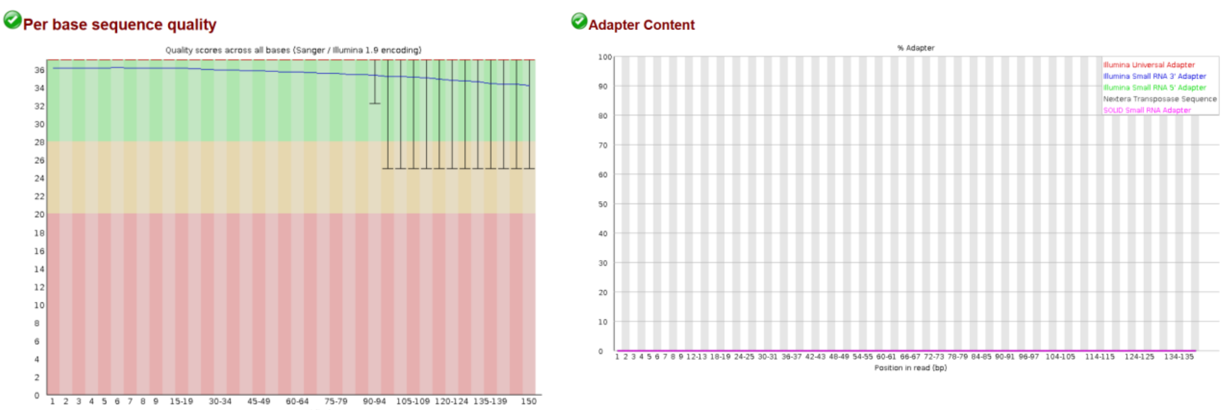

Figure 4. Per-base sequence quality and adapter content of reverse reads from the *Saccharomyces cerevisiae* mutant sample

Alignment statistics showed that both wild-type reads (Table 1) and mutant reads (Table 2) had good mapping performance. Therefore, the alignment results were considered suitable for downstream variant calling and annotation steps.

Table 1. Alignment and coverage statistics of the *Saccharomyces cerevisiae* wild-type sample

Parameter	Result	Description
Total primary reads	8,104,002	
Mapped reads	7,670,785 (94.65%)	>90% = good
Properly paired reads	7,561,448 (93.31%)	>90% = good
Nuclear genome coverage	99.13%	>95% = good
Mean nuclear depth	91.06x	>30x = good
Mean mapping quality	53.72	>30 = good

Table 2. Alignment and coverage statistics of the *Saccharomyces cerevisiae* mutant sample

Parameter	Result	Description
Total primary reads	7,137,298	
Mapped reads	6,701,110 (93.89%)	>90% = good
Properly paired reads	6,602,624 (92.51%)	>90% = good
Nuclear genome coverage	98.89%	>95% = good
Mean nuclear depth	80.69x	>30x = good
Mean mapping quality	54.87	>30 = good

SnpEff statistics showed that variants found only in the mutant sample (mutant-specific variants) consisted of 36,871 variants, with a variant rate of one variant every 329 bp. Based on predicted impact, 304 effects (0.119%) were classified as HIGH impact and 7,218 effects (2.831%) were classified as MODERATE impact. The most frequent predicted effects were upstream and downstream gene variants, while variants with potentially stronger functional consequences, such as missense and frameshift variants, were also among the prominent effects (Figure 5).

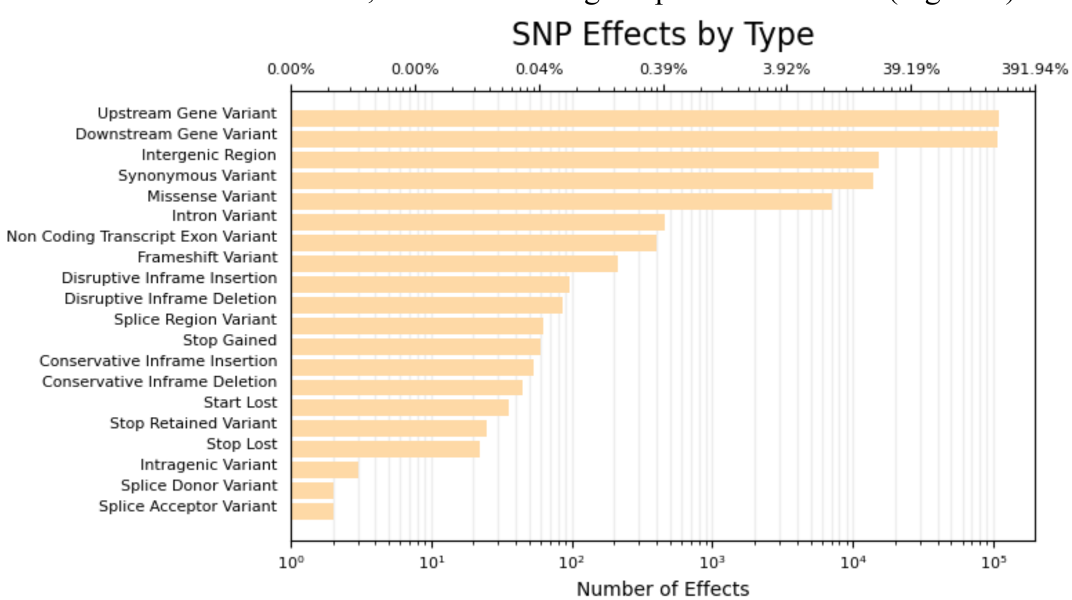

Figure 5. Predicted SNP effects in *Saccharomyces cerevisiae* mutant-specific variants

Discussion

Lu et al. (2023), as the source article of the data used in this reanalysis, discussed 61 genes related to ethanol production, sugar utilization, and associated metabolic pathways. In this WGS reanalysis, 2,794 genes were found to contain mutant-specific variants. Among these genes, 27 overlapped with the ethanol-related genes discussed by Lu et al. (2023). From these overlapping genes, MAL31, AAD4, and NTH1 were prioritized as candidate genes potentially associated with the higher ethanol production phenotype, based on their predicted variant impact, predicted effect type, and relevance to ethanol-related biological processes (Table 3).

Table 3. Top three prioritized candidate genes potentially contributing to higher ethanol production.

Gene	Impact	Predicted effect	Potential relevance
MAL31	3 HIGH; 16 MODERATE	2 frameshift; 1 frameshift and missense; 16 missense	Sugar transport and utilization
AAD4	1 HIGH; 6 MODERATE	1 stop-gained; 6 missense	Alcohol and aldehyde metabolism
NTH1	1 HIGH; 1 MODERATE	1 frameshift; 1 missense	Trehalose metabolism, stress response, and carbon reserve

To further evaluate the potential roles of these genes, functional validation is still required because WGS analysis alone does not directly test the biological effects of the identified variants. Additional analyses, such as gene expression analysis, targeted mutation confirmation, or fermentation-related assays, would help determine whether MAL31, AAD4, and NTH1 contribute to the higher ethanol production phenotype.

Conclusion

This WGS reanalysis identified variants specific to the high-ethanol-producing *Saccharomyces cerevisiae* mutant sample compared with the wild-type sample. The results highlight MAL31, AAD4, and NTH1 as the strongest candidate genes based on their predicted variant impact, predicted effect type, and relevance to ethanol-related biological processes. These genes are involved in sugar transport and utilization, alcohol and aldehyde metabolism, and trehalose-related stress or carbon reserve pathways, suggesting their potential contribution to the higher ethanol production phenotype.

References

Koboldt, D. C., 2020, Best practices for variant calling in clinical sequencing, *Genome Medicine*, 12, 91.

Lu, Z., Guo, L., Chen, X., Lu, Q., Wu, Y., Chen, D., Wu, R., & Chen, Y., 2023, Omics Sequencing of *Saccharomyces cerevisiae* Strain with High Ethanol Yield and Analysis of Metabolism, *Fermentation*, 9(5), 483.

NCBI Reference Genome: GCF_000146045.2_R64, *Saccharomyces cerevisiae* S288C genome assembly R64, https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_000146045.2/

NCBI SRA data: wild-type sample = SRR21747981; mutant sample = SRR21747980.

Parapouli, M., Vasileiadis, A., Afendra, A. S., & Hatziloukas, E., 2020, *Saccharomyces cerevisiae* and its industrial applications, *AIMS Microbiology*, 6(1), 1–31.

SnpEff database: R64-1-1.99.

Tools: SRA Toolkit v3.4.1, FastQC v0.11.8, fastp v1.3.3, BWA-MEM v0.7.19-r1273, SAMtools v1.23.1, BCFtools v1.19, FreeBayes v1.3.8, SnpEff/SnpSift v5.4c, and ngi_visualizations.